MAT 201 - MATHEMATICAL METHODS I

LECTURE 10: MULTIPLE INTEGRALS

LECTURE C24-1-10 :: MON., FEB. 17, 2025 8:00 AM (WAT) - FRI., FEB. 21, 2025 6:00 PM (WAT)

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OUTLINE

- Introduction
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    - Volumes
- Double Integral
- Triple Integral
- Summary
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INTRODUCTION

Area under a curve

Let *f* be a continuous function of one variable defined on the closed interval $[a, b]$ and supposed that *f* has only nonnegative values, then the definite integral

$$\int_a^b f(x)dx \tag{1}$$

exists and gives the area under the curve $f(x)$.

INTRODUCTION

Volume under a graph

Let *R* be the rectangle $\{(x, y)|a \leq x \leq b, c \leq y \leq d\}$ and let *f* be continuous and nonnegative on *R*. Then the volume *V* under the graph of *f* over *R* is

$$\int_a^b \int_c^d f(x, y)dydx = \int_c^d \int_a^b f(x, y)dxdy \tag{2}$$

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EXAMPLE 1

Calculate the volume under the graph of $z = 4 - x^2 - y^2$ over the square

$$R = \{(x, y)| - 1 \leq x \leq 1, -1 \leq y \leq 1\} \tag{3}$$

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EXAMPLE 1 [CONT.]

$$
\begin{aligned}
V &= \int_{-1}^{1}\int_{-1}^{1}(4-x^2-y^2)dydx && (4)\\
&= \int_{-1}^{1}\left(4y-x^2y-\frac{1}{3}y^2\right)\Bigg|_{y=-1}^{y=1}dx && (5)\\
&= \int_{-1}^{1}\left(\left(4-x^2-\frac{1}{3}\right)-\left(-4+x^2+\frac{1}{3}\right)\right)dx && (6)\\
&= \int_{-1}^{1}\left(8-2x^2-\frac{2}{3}\right)dx=\frac{40}{3} && (7)
\end{aligned}
$$

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DOUBLE INTEGRAL

The double integral of *f* on *R*, denoted by $\iint_R f dA$ (or $\iint_R f(x,y)dA$ or $\iint_R f(x,y)dxdy$), is the limit of the Riemann sum S as the dimensions $\triangle\ x_i$ and $\triangle\ y_j$ of the subrectangles R_{ij} all approach zero, that is

$$\iint_R f dA = \lim_{\forall \triangle x_i, \triangle y_j \to 0} \sum_{i,j=1}^{n} f(c_{ij})\ \triangle\ x_i\ \triangle\ y_j, \tag{8}$$

provided, of course, that this limit exists. c_{ij} is any point in the subrectangle $R_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j]$. When $\iint_R f dA$ exists, we say that *f* is integrable on *R*.

DOUBLE INTEGRAL

Fubini's theorem

If f is continuous on the closed rectangle R, then $\iint_R f dA$ exists.

$$\iint_R f dA = \int_a^b \int_c^d f(x,y) dy dx = \int_c^d \int_a^b f(x,y) dx dy \tag{9}$$

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PROPERTIES OF DOUBLE INTEGRAL

Suppose that f and g are both integrable on the closed rectangle R, then

$$\iint_R (f+g)dA = \iint_R fdA + \iint_R gdA \tag{10}$$

$$\iint_R cfdA = c\iint_R fdA; \quad c \in \mathbb{R} \tag{11}$$

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PROPERTIES OF DOUBLE INTEGRAL

Suppose that f and g are both integrable on the closed rectangle R, then

$$\iint_R f(x,y)dA \leq \iint_R g(x,y)dA; \quad f(x,y) \leq g(x,y) \tag{12}$$

$$\left|\iint_R f dA\right| \leq \iint_R |f| dA \tag{13}$$

EXAMPLE 2

Evaluate the double integral $\int_0^1 \int_{e^y}^y dxdy$.

$$\int_0^1 \int_{e^y}^y dxdy = \int_0^1 x|_{e^y}^e dy \tag{14}$$

$$= \int_0^1 (e - e^y)\, dy \tag{15}$$

$$= 1 \tag{16}$$

TRIPLE INTEGRAL

The triple integral of f on B, denoted by $\iiint_B f dV$ (or $\iiint_B f(x,y,z)dV$ or $\iiint_B f(x,y,z)dxdydz$), is the limit of the Riemann sum S as the dimensions $\triangle\ x_i$, $\triangle\ y_j$ and $\triangle\ z_k$ of the subboxes B_{ijk} all approach zero, that is

$$\iiint_B f dV = \lim_{\forall \triangle x_i, \triangle y_j, \triangle z_k \to 0} \sum_{i,j,k=1}^{n} f(c_{ijk})\ \triangle\ x_i\ \triangle\ y_j\ \triangle\ z_k, \tag{17}$$

provided that this limit exists. c_{ijk} is any point in the subbox $B_{ijk} = [x_{i-1}, x_i] \times [y_{j-1}, y_j] \times [z_{k-1}, z_k]$. When $\iiint_R f dV$ exists, we say that f is integrable on B.

FUBINI'S THEOREM

$$\iiint_B f dV = \int_a^b \int_c^d \int_p^q f(x,y,z) dx dy dz \tag{18}$$

$$= \int_a^b \int_p^q \int_c^d f(x,y,z) dy dz dx \tag{19}$$

$$= \int_c^d \int_a^b \int_p^q f(x,y,z) dz dx dy \tag{20}$$

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FUBINI'S THEOREM

$$\iiint_B f dV = \int_c^d \int_p^q \int_a^b f(x,y,z) dx dz dy \tag{21}$$

$$= \int_p^q \int_a^b \int_c^d f(x,y,z) dy dx dz \tag{22}$$

$$= \int_p^q \int_c^d \int_a^b f(x,y,z) dx dy dz \tag{23}$$

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EXAMPLE 3

Let $B = [-2, 3] \times [0, 1] \times [0, 5]$ and $f(x, y, z) = x^2 e^y + xyz$, evaluate

$$I = \iiint_B \left(x^2 e^y + xyz\right) dV \tag{24}$$

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EXAMPLE 3 [CONT.]

$$\iiint_B \left(x^2e^y + xyz\right) dV = \int_{-2}^{3}\int_{0}^{1}\int_{0}^{5} \left(x^2e^y + xyz\right) dzdydx \tag{25}$$

$$= \int_{-2}^{3}\int_{0}^{1} \left(x^2e^yz + \frac{1}{2}xyz^2\right)\Big|_{z=0}^{z=5} dydx \tag{26}$$

$$= \int_{-2}^{3}\int_{0}^{1} \left(5x^2e^y + \frac{25}{2}xy\right) dydx \tag{27}$$

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EXAMPLE 3 [CONT.]

$$\iiint_B \left(x^2e^y + xyz\right) dV = \int_{-2}^{3} \left(5x^2e^y + \frac{25}{2}xy^2\right)\Big|_{y=0}^{y=1} dx \tag{28}$$

$$= \int_{-2}^{3} \left(5(e-1)x^2 + \frac{25}{2}x\right) dx \tag{29}$$

$$= \frac{175}{3}(e-1) + \frac{125}{8} \tag{30}$$

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SUMMARY

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